

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12415065
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Hsu; Feng-Hsiung et al.

Artificial retinal prosthesis for providing color visual perception

Abstract

An artificial retinal prosthesis is disclosed, which comprises a plurality of pixel group units to output a spatiotemporal electrical stimulation for inducing color perception. Each of the pixel group units comprises a main pixel unit and at least one surrounding pixel unit. The main pixel unit and the surrounding pixel unit respectively outputs an electrical stimulation waveform according to a first stimulation cycle and a second stimulation cycle. Both of the first stimulation cycle and the second stimulation cycle have a first-half duration and a second-half duration. The first-half duration of the first stimulation cycle has inactive period greater than 20% and less than 80% and the rest of the first stimulation cycle is active period. The second-half duration of the first stimulation cycle is inactive period. The first-half duration of the second stimulation cycle is active period and the second-half duration of the second stimulation cycle is inactive period.

Inventors: Hsu; Feng-Hsiung (Hsinchu, TW), Chen; Yung-Chan (Hsinchu, TW), Fan; Long-Sheng (Hsinchu, TW), Lin; Lee (Hsinchu, TW)

Applicant: IRIDIUM MEDICAL TECHNOLOGY CO., LTD. (Hsinchu, TW)

Family ID: 1000008819513

Assignee: IRIDIUM MEDICAL TECHNOLOGY CO., LTD. (Hsin-Chu, TW)

Appl. No.: 17/704815

Filed: March 25, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220211999 A1	Jul. 07, 2022

Related U.S. Application Data

continuation-in-part parent-doc US 16230218 20181221 ABANDONED child-doc US 17704815

Publication Classification**Int. Cl.:** A61N1/05 (20060101)**U.S. Cl.:**

CPC A61N1/0543 (20130101);

Field of Classification Search**USPC:** None

References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6804560	12/2003	Nisch et al.	N/A	N/A
7751896	12/2009	Graf et al.	N/A	N/A
9155881	12/2014	Fan	N/A	N/A
2008/0294224	12/2007	Greenberg	607/54	A61N 1/0543
2010/0211168	12/2009	Goertz et al.	N/A	N/A
2011/0202132	12/2010	Meijer	N/A	N/A
2014/0121738	12/2013	Arafune	607/116	A61N 1/0543
2015/0049231	12/2014	Chen et al.	N/A	N/A
2016/0038739	12/2015	Liu et al.	N/A	N/A
2016/0118429	12/2015	Otsuji	N/A	N/A
2016/0256677	12/2015	Song et al.	N/A	N/A
2017/0224998	12/2016	Gefen	N/A	A61F 9/00

Primary Examiner: Lee; Erica S*Attorney, Agent or Firm:* MUNCY, GEISSLER, OLDS & LOWE, P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application is a continuation-in-part application of U.S. patent application Ser. No. 16/230,218, filed on Dec. 21, 2018, which claims the benefit of U.S. Provisional Patent Application No. 62/610,004, entitled “System for Artificial Retina Prosthesis,” which was filed on Dec. 22, 2017, and the disclosure of which is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

(1) The present invention relates to an artificial retinal prosthesis, and more particularly to an artificial retinal prosthesis for providing color visual perception.

BACKGROUND OF THE INVENTION

(2) Currently, among the patients with visual deterioration, some patients choose to implant an

artificial retina to improve their vision. At present, expensive artificial retinas of the commercial standard with low pixels have a limited improvement on the quality of life of patients. In view of this, many companies as well as academic and research institutes have begun to actively invest in the improvement of microsystem for artificial retina.

(3) In order to give the users a more comfortable visual experience, many R&D teams are actively making improvements on the image resolution. For example, U.S. Pat. No. 7,751,896 B2, No. U.S. Pat. No. 6,804,560 B2 improve the signal transmission in the artificial retina by adding components such as an amplifier or a photosensitive reference component to the circuit, so that the electrical stimulation signals of the artificial retina are more even, when the patient wears the above artificial retina, it is just like the response of eyes to ambient light conditions under natural conditions. There are also other teams that focus on the colors of the image, hoping to upgrade the conventional artificial retinas that only show black and white images to color images. For example, in U.S. Pat. No. 7,840,274 B2, the artificial retina comprises a color image receiver for receiving a color image and converting the color image into an electrical signal, and an image processing unit coupled to the color image receiver for processing the electrical signal. In the patent, a plurality of pixel electrodes are driven by data from the image processing unit to stimulate the optic nerve by time mode to produce a perception of color images. As far as we know, however, there is no published evidence that the time mode stimulation scheme as described in said patent works universally, reliably, or at all.

(4) At present, a number of pixel units of the artificial retina continues to increase, which has been greatly advanced for artificial retinas having only a few tens of pixel units in the past. In contrast, related researches on artificial retinal systems that provide color visual perception are still at a very early stage, and even though many manufacturers and teams have proposed various artificial retinal systems that provide color visual perception, there is no corresponding product/system that has been manufactured, or it is not good enough to achieve color visual perception after practical operations. Obviously, there is still a lot of room for development in developing artificial retina systems providing color visual perception, depending on the continuous investment and improvement of relevant teams.

SUMMARY OF THE INVENTION

(5) The present invention is an artificial retinal prosthesis to electrically stimulate a retina of an eye by a spatiotemporal electrical stimulation. The artificial retinal prosthesis comprises a plurality of pixel group units arranged in an array for inducing perception of different colors. Each of the pixel group units comprises a main pixel unit and at least one surrounding pixel unit adjacent to the main pixel unit, which are configured to receive an external visual image entering eyes of the user. The main pixel unit and the surrounding pixel unit respectively outputs an electrical stimulation waveform according to a first stimulation cycle and a second stimulation cycle respectively to retinal cells of the user for inducing a color perception. The first stimulation cycle of the main pixel unit has a first-half duration and a second-half duration after the first-half duration, the first-half duration of the first stimulation cycle has inactive period greater than 20% and less than 80% and the rest of the first stimulation cycle is active period, and the second-half duration of the first stimulation cycle is inactive period. The second stimulation cycle of the surrounding pixel unit has a first-half duration and a second-half duration after the first-half duration, the first-half duration of the second stimulation cycle is active period and the second-half duration of the second stimulation cycle is inactive period.

(6) The system for artificial retinal prosthesis of the present invention causes the stimulation of the pixel electrodes and the spectrum of the external visual image entering the user's eyes to change synchronously along with different time sequences and different spatiotemporal distributions of each retinal cell, thereby stimulating the patient's retinal cells to provide the patient with a color image perception that assists the patient in truly obtaining RGB color vision. The spatiotemporal stimulation creates color perception in essential the same way as the so-called Fechner Color effect.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram of the operation of a system for artificial retinal prosthesis with color vision according to an embodiment of the present invention;
- (2) FIG. 2 is a color shutter structure according to an embodiment of the present invention;
- (3) FIG. 3 is a schematic view showing an electric field applied to the color shutter structure of FIG. 2 of the present invention;
- (4) FIG. 4 is a schematic diagram of activation of row-to-row of the system for artificial retinal prosthesis with color vision according to an embodiment of the present invention;
- (5) FIG. 5 is a schematic view showing the states in which pixel electrodes are turned on and off according to embodiment 3 of the present invention;
- (6) FIG. 6 is a schematic view showing the states in which the pixel electrodes are turned on and off according to embodiment 4 of the present invention;
- (7) FIG. 7 illustrates an artificial retinal prosthesis according to an exemplary embodiment of the invention;
- (8) FIG. 8 illustrates a pixel group unit with stimulation cycle according to an exemplary embodiment of the invention; and
- (9) FIG. 9 illustrates various stimulation cycle for the main pixel unit and the surrounding pixel unit according to an exemplary embodiment of the invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (10) Referring to FIG. 1. A system for artificial retinal prosthesis with color vision in an embodiment of the present invention mainly comprises an artificial retinal prosthesis **10** and a color shutter **20**, and the color shutter **20** is fitted on a goggle **30**. In other embodiments, the color shutter **20** can also be fitted to a pair of glasses or other devices that can be worn by a user.
- (11) The artificial retinal prosthesis **10** can send a wireless signal to the goggle **30** to control the color shutter **20** of the goggle **30**. For example, when the artificial retinal prosthesis **10** needs a red light stimulus, the artificial retinal prosthesis **10** sends a wireless signal **S1** to the goggle **30** to activate a red color shutter in the color shutter **20**, so that only red light can pass through the red color shutter of the goggle **30** to reach the artificial retinal prosthesis **10**. If a blue light stimulus is required, the artificial retinal prosthesis **10** sends a wireless signal **S2** to the goggle **30** to activate a blue color shutter in the color shutter **20**, so that blue light can pass through the goggle **30** to reach the artificial retinal prosthesis **10**. Likewise, when a green light stimulus is required, the artificial retinal prosthesis **10** sends a wireless signal **S3** to the goggle **30** to activate a green color shutter, so that green light can pass through the goggle **30** to reach the artificial retinal prosthesis **10**. Subsequently, after the artificial retinal prosthesis **10** receives a specific incident light such as red light, blue light, or green light through the color shutter **20**, a pixel electrode array in the artificial retinal prosthesis **10** is electrically stimulated by a spatiotemporal electrical stimulation.
- (12) When the pixel electrodes in the artificial retinal prosthesis **10** are defined as P_{xy} according to the spatial positions, such as **P11**, **P12**, **P13**, **P22**, **P23**, the above-mentioned “spatiotemporal electrical stimulation” refers to different stimulations given to corresponding optic nerves by different P_{xy} at different times, for example, **P11** and **P12** stimulations are given at time point **t1**, but the remaining pixel electrodes are not.
- (13) The artificial retinal prosthesis **10** is disposed on the retina of the eye structure, and can be disposed on the sub-retina or the epi-retina as needed in actual use without particular limitation. This embodiment is disposed on the sub-retina. The artificial retinal prosthesis **10** comprises a plurality of pixel arrays and a processing module disposed correspondingly to the plurality of pixel arrays. Each of the plurality of pixel arrays comprises a substrate and a plurality of sub-pixels disposed on the substrate for receiving a color image. In this embodiment, the substrate can be a

thin flexible silicon substrate that can be deformed and bent as desired, so that it can be bent as much as possible into a structure conforming to the shape of a human eye and disposed in the eye of a patient.

(14) In actual manufacturing, for example, the substrate can be fabricated based on a manufacturing process using a Silicon On Insulator (SOI) chip, and formed by thinning the chip after a Metal-Oxide-semiconductor (MOS) fabrication. The processing module can include a correlated double sampling unit (CDS), an analog-to-digital converter (ADC), a digital core, and a digital-to-analog converter (DAC) to process a signal of the pixel array. However, the components included in the processing module are not limited to the above components, technicians of this field can add or delete based on actual needs and designs.

(15) Each of the plurality of sub-pixels comprises at least one pixel electrode, a photodiode, and a circuit architecture electrically connected to the photodiode. After an incident light emit to the photodiode, the incident light is converted into an electric charge and a photovoltaic potential, and a light-induced electrical stimulation signal is generated according to an intensity ratio of the incident light. The light-induced electrical stimulation signal generates the spatiotemporal electrical stimulation to stimulate the patient's retinal cells, thereby producing a color image.

(16) It should be additionally explained that, in another embodiment of the present invention, the color shutter **20** may not be assembled on the goggle **30**, but can be integrated into a single structure with the artificial retinal prosthesis **10**. That is, the color shutter **20** can be formed on the pixel array of the artificial retinal prosthesis **10** and can include a plurality of optical shutter units corresponding to different colors. For example, the color shutter **20** can include red shutters formed in a first row to a third row of the pixel array, green shutters formed in a fourth row to a sixth row, and blue shutters formed in a seventh row to a ninth row.

(17) For one of the examples of the color shutter **20**, please refer to FIG. 2. The color shutter **20** can include a first substrate **21**, a second substrate **22** disposed oppositely to the first substrate **21**, an electrode **23** disposed between the first substrate **21** and the second substrate **22**, a hydrophobic layer **24** disposed between the electrode **23** and the second substrate **22**, a first fluid layer **25** disposed between the hydrophobic layer **24** and the second substrate **22**, and a second fluid layer **26** disposed between the hydrophobic layer **24** and the first fluid layer **25**, wherein the first fluid layer **25** and the second fluid layer **26** are immiscible with each other.

(18) In this embodiment, the first substrate **21** and the second substrate **22** are transparent and can be formed with the same or different materials, such as glass, resin, polycarbonate (PC), and the like.

(19) The first fluid layer **25** can be a conductive or polarized water or salt solution; and the second fluid layer **26** can be an oily medium, so that when the first fluid layer **25** and the second fluid layer **26** coexist between the second substrate **22** and the hydrophobic layer **24**, a two-layer structure can be formed without being miscible. In this embodiment, the second fluid layer **26** can be a mixture of oils with different colors, such as can be selected from a green oil, a red oil, a blue oil, or any combinations of the above oils.

(20) The hydrophobic layer **24** can be a functional layer with low surface energy and high stability, and specifically, can be made of a polymer or a silicon dioxide layer. For example, the polymer used for the hydrophobic layer **24** may be a fluoropolymer such as Cytop or amorphous Teflon, or a hydrocarbon polymer may also be used. If silicon dioxide is used, its surface needs to be treated hydrophobically.

(21) The electrode **23** is disposed on the first substrate **21** to apply a voltage to the first fluid layer **25**. The electrode **23** used in the present embodiment is preferably a transparent electrode made of any suitable conductive material such as indium tin oxide (ITO). The above is merely illustrative, and the present invention is not limited thereto, and the color shutter **20** may employ other devices such as a light filter.

(22) In another embodiment of the present invention, the color shutter **20** can further include an

optical sensor for sensing ambient light and/or a variable light filter for automatically controlling light passing through the color shutter **20** according to environmental conditions.

(23) The principle used by the color shutter **20** is an electrowetting effect, that is, a wettability of the oily medium on the substrate is controlled by changing a voltage between the oily medium and the hydrophobic layer **24** (insulating layer). More specifically, the oily medium is deformed and displaced by changing a contact angle. The term “wetting” used above refers to the process of a fluid on a solid surface being replaced by another fluid. The fluid on the solid surface (i.e., the hydrophobic layer **24**) can diffuse, at this time, the adhesion of the fluid on the solid surface is greater than the cohesion, referred to as “wetting.” Conversely, when the fluid on the solid surface (i.e., the hydrophobic insulating layer) cannot diffuse, the contact surface has a tendency to shrink into a spherical shape, which is called “non-wetting”, and “non-wetting” refers to the adhesion of the fluid on the solid surface being smaller than the cohesion.

(24) Returning to the present invention, the first fluid layer **25** and the second fluid layer **26** are immiscible with each other without applying an electric field to the fluids (closed state) to form a two-layer structure in which the first fluid layer **25** is diffused to form as a fluid layer adjacent to the second substrate **22**; and the second fluid layer **26** also diffuses to form a fluid layer adjacent to the hydrophobic layer **24** and serves as color pixels. However, when an electric field is applied to the fluids (on state), the second fluid layer **26** is broken into small droplets to cause the color shutter **20** to exhibit a transparent color, as shown in FIG. 3.

(25) Therefore, in order to obtain various display results, the second fluid layer **26** (i.e., the oily medium) can be designed to have a desired color, and a surface of the oily fluid can be controlled to change the pixels by controlling the voltage.

(26) In the other embodiment, the anisotropic color pigment particles (say pigment needles) in fluid suspensions could be utilized in an alternative color shutter. Three shutters in tandem, with Yellow, Cyan, and Magenta color pigments, would be needed. Each color shutter would be turned on by applying sufficient large voltage across the fluid to align the particle with the field. Alternatively, another type of color shutter with electrophoretic cells in shutter mode could be used. This is somewhat harder to reach adequate speed, but can work with optimized cells.

(27) When the system for artificial retinal prosthesis with color vision of this embodiment is in use, the color image is converted into the light-induced electrical stimulation signal by the photodiode of the sub-pixel, and the spatiotemporal electrical stimulation is generated to provide the patient with color perception. As a specific example, the spatiotemporal electrical stimulation of about 4 Hz to 8 Hz (preferably 7 Hz) can be divided into seven equally spaced phases within one cycle, producing color sensations of red (R), green (G) and blue (B).

(28) Further explain how to provide the patient with color perception by the spatiotemporal electrical stimulation as below.

Embodiment 1

(29) In this embodiment, the pixel electrodes in each of the pixel arrays are arranged in 1 column of 9 rows and classified into three groups corresponding to the specific color perceptions. A time series takes 7 equally spaced frames as a cycle, and the cycles per second (cps) can be between 7 and 8, so the frames per second (fps) are between 49 and 56, and the cycle between two of the frames is approximately 20 ms.

(30) Please refer to Table 1, wherein “_” means the pixel electrode is turned off and “|” means the pixel electrode is turned on.

(31) TABLE-US-00001

TABLE 1	Frame	1	2	3	4	5	6	7	R	Row	1	—	—	—		Row	2	—	—	—	—	—		Row	3	—	—	—		G	Row	4	—	—	—		Row	5	—	—	—	—		—	Row	6	—	—	—		B	Row	7	—	—	—		Row	8	—	—	—		—	—	Row	9	—	—	—	
---------	-------	---	---	---	---	---	---	---	---	-----	---	---	---	---	--	-----	---	---	---	---	---	---	--	-----	---	---	---	---	--	---	-----	---	---	---	---	--	-----	---	---	---	---	---	--	---	-----	---	---	---	---	--	---	-----	---	---	---	---	--	-----	---	---	---	---	--	---	---	-----	---	---	---	---	--

(32) For the 3 rows of R/G/B strips, the span is 240 μm (30 m*8) strip width.

(33) Row 1 to row 9 start rolling at the same time. It can be found from Table 1 that all the pixel electrodes are turned off in frame 1; the pixel electrodes of row 1, row 3 to row 9 are turned on

while the other pixel electrodes are turned off in frame 5; while in frame 7, all the pixel electrodes are turned on except for the pixel electrodes of row 5 and row 8 being turned off. Based on the arrangement and operation of the pixel electrodes described above, the patient can perceive colors on the corresponding pixel electrodes, for example, in the cycles from frame 6 to frame 7, the patient can perceive red in the pixel electrodes of row 2.

(34) If power attenuation problem is taken into consideration, in other embodiments, electrical stimulations of the above-mentioned “rows” are not simultaneously sent out in the same frame. If all the “rows” in the same frame are enabled at exactly the same time, the artificial retinal prosthesis **10** will consume a very large amount of power and cause a drop in power, even making the artificial retinal prosthesis **10** unable to function properly. In order to avoid the above problem, in the cycles of the same frame, when the state of the pixel electrodes is “|” representing being turned-on, they will be activated row-to-row. That is to say, electrical stimulations of the subsequent rows will slightly lag behind the previous pixel electrode; however, when the state of the pixel electrodes is “-” representing being turned-off, as in the first column to the third column (frame 1 to frame 3) of Table 1 above, the pixel electrodes in the columns cannot be activated and electrical stimulations are not sent out from the columns. Please refer to FIG. 4, where the horizontal axis (x-axis) is time and the vertical axis (y-axis) is electrical signal strength, that is, voltage. “Pulse width” in FIG. 4 refers to pulse duration, and “Interval” represents time delay. In Tables 1 and 2, the time for turning on each of the rows is simultaneous, but in reality, there may be a time difference between turning on each of the rows (such as the time delay of ns level), for example, after row 7 is turned on and off, then it is the turn for row 8 to be turned on and off.

Embodiment 2

(35) Please refer to Table 2. In this embodiment, the pixel electrodes are arranged in 1 column of 6 rows, and the electrodes are classified into three groups with each of the groups respectively corresponding to a specific color. The setting of the time series in this embodiment is the same as that of Embodiment 1.

(36) TABLE-US-00002 TABLE 2 Frame 1 2 3 4 5 6 7 R Row 1 — — — — — || Row 2 — — — — — |
||| G Row 3 — — — — — || — Row 4 — — — — — ||| B Row 5 — — — — — || — Row 6 — — — — — |||
|

(37) This embodiment is similar to that of Table 1, the pixel electrodes of row 1 to row 6 are all turned off in frame 1; the pixel electrodes of row 2 to row 6 are all turned on in frame 5, only the pixel electrodes of row 1 are turned off; while in frame 7, all the pixel electrodes are turned on except for the pixel electrodes of row 3 and row 5 being turned off.

Embodiment 3

(38) Please refer to FIG. 5, wherein “D” represents a dummy electrode, “R” represents an electrode group that can perceive red correspondingly, “G” represents an electrode group that can perceive green correspondingly, and “B” represents an electrode group that can perceive blue correspondingly. In this embodiment, the pixel electrodes are arranged in 7 rows of 6 columns, and each of the R, G, B pixel electrodes is surrounded by 9 dummy electrodes.

(39) If row 2 of column 3 is taken as an example, all the pixel electrodes are turned off in frame 1 to frame 4, and the surrounding dummy electrodes are also turned off; while the pixel electrodes are turned on in frame 5 to frame 6, so only green light can pass through the goggle **30** to reach the artificial retinal prosthesis **10**, and at the same time, the surrounding dummy electrodes are also turned on. According to the operation of the pixel electrodes described above, the patient can have a green visual perception in the pixel electrodes of column 3 and row 2, and by the above arrangement of the dummy electrodes, a visual contrast can be generated between the pixel electrodes corresponding to the specific colors and the surrounding areas thereof, and the effect of enhancing the color perception of the patient is achieved.

Embodiment 4

(40) Referring to FIG. 6, “D” represents a dummy electrode, “R” represents an electrode group that

— D Row 60 — — — — ||| — D Row 61 ||| — — — — B Row
 62 || — — — — D Row 63 ||| — — — — D Row 64 — — — —
 — — — — ||| R Row 65 — — — — || D Row 66 — — — — ||
 || D Row 67 — — — — ||| — — — — G Row 68 — — — — || — — — — D Row 69
 — — — — ||| — — — — D Row 70 ||| — — — — B Row 71 || — — — —
 — — — — D Row 72 ||| — — — — D Row 73 — — — — ||| R
 Row 74 — — — — || D Row 75 — — — — ||| D Row 76 — — — —
 — — ||| — — — — G Row 77 — — — — || — — — — D Row 78 — — — — ||| —
 — — — — D Row 79 ||| — — — — B Row 80 || — — — — D
 Row 81 ||| — — — — D Row 82 — — — — ||| R Row 83 — — — —
 — — — — || D Row 84 — — — — ||| D Row 85 — — — — ||| — —
 — — G Row 86 — — — — || — — — — D Row 87 — — — — ||| — — — — D Row
 88 ||| — — — — B Row 89 || — — — — D Row 90 ||| — — — —
 — — — — D Row 91 — — — — ||| R Row 92 — — — —
 — || D Row 93 — — — — ||| D Row 94 — — — — ||| — — — — G Row 95 —
 — — — — || — — — — D Row 96 — — — — ||| — — — — D Row 97 ||| — — — —
 — — — — B Row 98 || — — — — D Row 99 ||| — — — — D
 Row 100 — — — — ||| R Row 101 — — — — || D Row 102 —
 — — — — ||| D Row 103 — — — — ||| — — — — G Row 104 — — — — ||
 — — — — D Row 105 — — — — ||| — — — — D Row 106 ||| — — — —
 B Row 107 || — — — — D Row 108 ||| — — — — D Row 109
 — — — — ||| R Row 110 — — — — || D Row 111 — — — —
 — — — — ||| D Row 112 — — — — ||| — — — — G Row 113 — — — — || — — — —
 — — D Row 114 — — — — ||| — — — — D Row 115 ||| — — — — B Row
 116 || — — — — D Row 117 ||| — — — — D Row 118 — — — —
 — — — — ||| R Row 119 — — — — || D Row 120 — — — —
 — — ||| D Row 121 — — — — ||| — — — — G Row 122 — — — — || — — — —
 D Row 123 — — — — ||| — — — — D Row 124 ||| — — — — B Row 125 ||
 — — — — D Row 126 ||| — — — — D Row 127 — — — —
 — — — — ||| R Row 128 — — — — || D Row 129 — — — —
 ||| D Row 130 — — — — ||| — — — — G Row 131 — — — — || — — — — D Row
 132 — — — — ||| — — — —

(45) FIG. 7 illustrates an artificial retinal prosthesis according to an exemplary embodiment of the invention. The artificial retinal prosthesis is configured to electrically stimulate a retina of an eye by a spatiotemporal electrical stimulation. The details of the spatiotemporal electrical stimulation will be described below.

(46) The artificial retinal prosthesis includes a plurality of pixel group units **30A**, **30B**, **30C** arranged in an array. The pixel group units **30A**, **30B**, **30C** are configured to receive an external visual image entering eyes of the user and for inducing perception of different colors.

(47) The pixel group unit **30A** comprises a main pixel unit **31A** and at least one surrounding pixel unit **32A** adjacent to the main pixel unit **30A**. The pixel group unit **30B** comprises a main pixel unit **31B** and at least one surrounding pixel unit **32B** adjacent to the main pixel unit **30B**. The pixel group unit **30C** comprises a main pixel unit **31C** and at least one surrounding pixel unit **32C** adjacent to the main pixel unit **30C**. In this embodiment, the pixel unit is formed in a hexagonal shape. In other embodiments, the pixel unit may be formed in a substantially elongated, circular, elliptic, triangular, quadrangular, pentagonal, heptagonal or octagonal shape.

(48) In this embodiment, the first closest pixel unit to the main pixel is defined as the surrounding pixel unit. In other embodiments, the Nth closest pixel unit (e.g., N=2, 3, 4, 5, 6 . . .) to the main pixel may be defined as the surrounding pixel unit.

(49) FIG. 8 illustrates a pixel group unit with stimulation cycle according to an exemplary

embodiment of the invention. Taking the pixel group unit **30A** as the example. The main pixel unit **31A** and the surrounding pixel units **32A** at different locations respectively output an electrical stimulation waveform according to different stimulation cycles to create a color perception.

(50) The main pixel unit **31A** and the surrounding pixel units **32A** output a first stimulation cycle and a second stimulation cycle respectively to retinal cells of the user. The first stimulation cycle of the main pixel unit **31A** has a first-half duration **311A** and a second-half duration **312A** after the first-half duration **311A**. The second stimulation cycle of the surrounding pixel unit **32A** has a first-half duration **321A** and a second-half duration **322A** after the first-half duration **321A**.

(51) The first-half duration **311A** of the first stimulation cycle has an inactive period **3111A** greater than 20% and less than 80% of the first-half duration **311A** and the rest of the first-half duration **311A** is an active period **3112A**, and the second-half duration **312A** of the first stimulation cycle is an inactive period **3121A**.

(52) The second stimulation cycle of each of the surrounding pixel units **32A** also has a first-half duration **321A** and a second-half duration **322A** after the first-half duration **321A**, all the first-half duration **321A** of the second stimulation cycle is active period and all the second-half duration **322A** of the second stimulation cycle is inactive period.

(53) In the above-mentioned, the active period refers to a period that electrode of the pixel unit outputs a stimulation to the retinal cells of the user and the inactive period refers to a period that electrode of the pixel unit does not output a stimulation to the retinal cells of the user.

(54) In the present invention, each of the pixel group units is configured to produce a specific color of vision perception. The specific color may be red, blue or green. The color of vision perception produced by the pixel group unit depends on a sequence and a length of active period and inactive period in the first-half duration of the first stimulation cycle of the main pixel unit.

(55) FIG. 9 illustrates various stimulation cycle for the main pixel unit and the surrounding pixel unit according to an exemplary embodiment of the invention. In an embodiment of the invention, a red color perception can be produced when the first-half duration **311A** of the main pixel unit **31A** is consisted of the inactive period **3111A** of approximately between 25% and 50% of the first-half duration **311A**, followed by the active period **3112A**. In an embodiment of the invention, a green color perception is produced when the first stimulation cycle **311A** of the main pixel unit **31A** is consisted of the active period **3112A** greater than 25% of the first-half duration **311A**, followed by the inactive period **3111A** of approximately between 30% and 80% of the first-half duration **311A**, followed by the active period **3112A**. In an embodiment of the invention, a blue color perception is produced when the first-half duration **311A** of the main pixel unit **31A** is consisted of the active period **3112A** of approximately between 20% and 80% of the first-half duration **311A**, followed by the inactive period **3111A**.

Claims

1. An artificial retinal prosthesis to electrically stimulate a retina of an eye by a spatiotemporal electrical stimulation, comprising: a plurality of pixel group units arranged in an array for inducing perception of different colors, each of the pixel group units comprising a main pixel unit and at least one surrounding pixel unit adjacent to the main pixel unit, which are configured to receive an external visual image entering eyes of the user; wherein the main pixel unit and the surrounding pixel unit respectively output an electrical stimulation waveform according to a first stimulation cycle and a second stimulation cycle respectively to retinal cells of the user for inducing a color perception; wherein the first stimulation cycle for the main pixel unit has a first-half duration and a second-half duration after the first-half duration, the first-half duration of the first stimulation cycle has an inactive period which is greater than 20% and less than 80% of the first-half duration, and the rest of the first-half duration is an active period, and the second-half duration of the first stimulation cycle is an inactive period; and wherein the second stimulation cycle for the

- surrounding pixel unit has a first-half duration and a second-half duration after the first-half duration, the first-half duration of the second stimulation cycle is an active period and the second-half duration of the second stimulation cycle is an inactive period.
2. The artificial retinal prosthesis as claimed in claim 1, wherein a red color perception is produced when the first-half duration of the main pixel unit consisted of the inactive period which is approximately between 25% and 50% of the first-half duration, followed by the active period.
 3. The artificial retinal prosthesis as claimed in claim 1, wherein a green color perception is produced when the first-half duration of the main pixel unit consisted of the active period which is greater than 25% of the first-half duration, followed by the inactive period which is approximately between 30% and 50% of the first-half duration, followed by the active period.
 4. The artificial retinal prosthesis as claimed in claim 1, wherein a blue color perception is produced when the first-half duration of the main pixel unit consisted of the active period which is approximately between 20% and 80% of the first-half duration, followed by the inactive period.
-